

PART 1 • THE 20-MINUTE EXPERIMENT

Zombie Invasion Prep • Sandbag Emergency Logistics

Student calculation packet • One copy per team, plus reference pages for each student

The challenge

With four workers, fill, close, carry and place as many acceptable sandbags as you safely can in 20 minutes. More empty bags may be supplied if the teacher has them. Completed means all four steps are finished at the marked destination. A partly filled bag or a bag still waiting to move does not count.

Before the clock starts

Put on work gloves and safety glasses. Listen to the teacher's demonstration and match the sample bag: about half to two-thirds full, flexible and comfortable to handle. Close it using the teacher's consistent demonstrated method. Real flood barriers often use folded/tucked tops rather than tied bags; this class uses the same closure for everyone. Do not pack bags solid or compete by making tiny bags.

Choose roles: hold/fill, shovel, close, move/place. You may change roles. Keep shovel swing zones and the carry path clear. Lift comfortably, use a buddy when needed, walk, and put bags down under control. Stop for pain, fatigue or unsafe conditions.

Timing and counting

Teacher gives one start signal and a hard stop at 20 minutes. Count only bags accepted at the destination at the stop. Work safely; this is a measurement, not a speed contest. Do not empty and refill a bag during the trial. Count each bag once.

If the supply runs out

Record the time you ran out and stop counting. Do not quietly report a full 20-minute maximum. A 20-minute count $\times$ 3 is still the observed output under the supply limit; it underestimates unrestricted production. Ask the teacher how to label and use this limited trial. If you produced zero, record the bottleneck; a zero rate cannot be used as a divisor.

Safety agreement

We work only with clean sand on a teacher-inspected dry practice site. We do not build live flood barriers or enter floodwater. We keep eyewear on, wear closed-toe footwear, avoid throwing sand and wash hands before eating. The teacher may stop the trial for weather or any hazard. Report an injury immediately.

Name / team: _____

Four team members: _____

Date / trial location: _____

Teacher-approved destination and carry route: _____

TEAM DATA & BASIC CALCULATIONS

Names: _____ Team: _____

Record what happened

Workers: 4 Trial duration: 20 minutes

Completed bags at final location, B: _____

Approximate one-way carrying distance: _____ ft

Shovels used: _____ Other moving equipment: _____

Was the full trial supplied with empty bags and sand? Yes / No

If not, when and what ran out? _____

Weather / ground / route conditions: _____

Closure / fill standard used: _____

Jobs at the start and any changes: _____

Main bottleneck and evidence (waiting, queues, crowded area): _____

1. Experimental team production rate

$R = B \times 3 = _____ \times 3 = _____ \text{ bags per team-hour}$

This means one team of four at the conditions you measured.

2. Experimental average per person

$R \div 4 = _____ \div 4 = _____ \text{ bags per person-hour}$

This is an average within the team, not a promise that a solo worker can do every job at this rate.

3. Fill represented by the completed bags

$B \div 75 = _____ \div 75 = _____ \text{ cubic yards of fill (estimate)}$

No weighing. We use the same 75 bags $\approx 1 \text{ yd}^3$ assumption throughout.

4. Sustained planning rates for Part 2

Normal: $S = R \times 0.75 = _____ \text{ bags per team-hour}$

Difficult conditions: $S = R \times 0.50 = _____ \text{ bags per team-hour}$

Keep rates to one decimal place, or retain calculator precision.

5. What would improve production?

One change and the resource it needs: _____

Why adding people alone might not help: _____

Data-quality note

Circle: Full trial / Supply-limited / Weather-stopped / Other

If invalid or zero, teacher assigns a labeled practice rate. Do not invent measured results.

COMMON REFERENCE • STRUCTURE ESTIMATES

These are classroom planning assumptions, NOT engineering specifications. Dimensions on scenario cards are proposed planning layouts, not permission to build. Adults and qualified responders approve real work.

Bag geometry model

One flexible filled bag occupies about 1 ft along a line, 0.5 ft across it and 0.25 ft (3 in) in height. The fill conversion remains 75 bags $\approx$ 1 yd³; geometry is a rough counting model, not a sand-volume measurement.

Single-row line: 1 bag per linear foot per layer.

Narrow line: 1 bag across; two layers = 2 bags per foot.

Broad barrier: triangular cross-section. Three layers have 3 bags across the bottom, 2 in the middle, 1 on top: 6 bags per foot; crest about 9 in. Four layers: 4 + 3 + 2 + 1 = 10 bags per foot; crest about 12 in.

Small erosion-control structure: 2 bags across $\times$ 2 layers = 4 bags per foot. These do not support slopes, roads or people.

Basin or low perimeter berm: 2 bags across $\times$ 2 layers = 4 bags per foot; crest about 0.5 ft. The oversized basin card explicitly requests 4 across $\times$ 4 layers = 16 per foot; crest only 1 ft.

Ground-cover anchor: one bag every 4 ft of perimeter, unless the card supplies a station count. Round bag count up.

Approved equipment station: stations $\times$ specified bags per station. These are inventory counts, not wind-load ratings.

How to calculate

Straight line: length $\times$ bags per foot. Add all separate lines.

U shape: front + two ends; do not count the open side.

L shape: add both legs. Rectangle perimeter = 2 $\times$ (length + width).

For basins, use the inside perimeter as a simple estimate; real corner and outer-row adjustments need extra bags. A labeled 10% contingency is acceptable if resources permit.

Round whole bags up. Base fill = bags $\div$ 75; report two decimals. Order fill to the next 0.25 yd³, while also checking on-site stock.

Example: a 20 ft broad three-layer line needs 20 $\times$ 6 = 120 bags; fill = 120 $\div$ 75 = 1.60 yd³; order 1.75 yd³ if buying in quarter yards.

Basin math • planning only

Volume (ft³) = inside length $\times$ inside width $\times$ water depth.

Gallons $\approx$ ft³ $\times$ 7.48. For water only, approximate load = gallons $\times$ 8.3 lb. Do not calculate bag weight.

Use only ground-level, flat, approved sites. Never rooftops, balconies or improvised raised platforms. A two-layer berm has a 0.5 ft theoretical crest; use a proposed operating depth no greater than 0.25 ft unless a qualified real-world design establishes otherwise. Leave freeboard (empty height), plan overflow, prevent access and label NON-POTABLE. Rainfall and liner integrity limit useful supply. This is not drinking or handwashing water.

COMMON REFERENCE • TIME & SUSTAINMENT

Same method for every student

R = your team's completed bags $\times 3$.

Use $S = 0.75R$ for normal planning, including short scenarios. Use $S = 0.50R$ for hot, steep, very muddy or otherwise difficult sustained work; explain your choice. Do not multiply both factors. These factors allow rest, water, role changes and routine delays; they do not replace separate delivery or transport analysis.

N = required bags; T = fully equipped four-person teams.

Experimental team-hours = $N \div R$.

Adjusted team-hours = $N \div S$.

Ideal elapsed production time = $N \div (T \times S)$.

Deadline minimum teams = round UP [$N \div (S \times \text{usable hours})$].

Usable hours = deadline minus initial wait for needed resources and any separate setup allowance. If usable hours ≤ 0 , current plan cannot meet the deadline.

Available teams = round DOWN (available workers $\div 4$). Partial groups can support delivery, traffic control or setup, but do not count as another measured team. Only count people actually assigned to the production chain; never assign them two simultaneous jobs.

Elapsed time is more than a division

Your measured rate already included closing, carrying and placing at the experimental route. Compare that distance and equipment with the card. Do not automatically add the same carrying time again. For a changed route, check transport capacity. With separate support workers, stages may overlap: chain rate is limited by the slowest stage. With shared workers, allow a sequential movement period or reduce effective team capacity and explain it. Add delivery waits, setup, inspections and finishing when not included. A capacity estimate is optimistic if people are shared.

A fast estimate: $\max(\text{adjusted production time, changed-route transport time}) + \text{unavoidable initial wait} + \text{setup/finish}$.

Use only if the overlapping stages are independently staffed. Otherwise draw a sequential schedule or reserve support workers first. Never double-count a person or cart.

Simplified classroom worker-support rules

Normal: plan 45 min working and 15 min resting/reorganizing each hour; $S = 0.75R$.

Difficult: plan 30 min working and 30 min resting/reorganizing each hour; $S = 0.50R$. Shade/cooling/warming breaks as appropriate. Stop if conditions become unsafe; a rate factor does not make unsafe work acceptable.

Drinking-water stock: plan 1 quart per assigned worker per elapsed hour, round up to whole gallons (4 quarts = 1 gallon), plus a 25% reserve rounded up. This is a stock-planning exercise, not a prescribed drinking dose; individuals drink as needed and supervisors adjust for conditions. Refill accessible coolers with safe drinking water.

Food: for more than 4 hours, plan a snack/meal break and one food serving per worker per 4-hour block, rounded up. Keep food away from sand and contamination.

Rotate lifting, shoveling and carrying; put rest areas off routes. Identify a first-aid adult, communication method and relief workers for long operations. Account for support personnel in water and food totals.

COMMON RESOURCE CATALOG

Catalog capacities are classroom assumptions, not vehicle specifications. Scenario cards give requests and site constraints; students calculate the resources to request. Catalog entries are things you can request, not free extras. State quantity, arrival time, operator and route.

People & PPE

Worker team: 4 equipped workers. Plan 2 shovels per active team as a starting allocation; fewer requires a rate reduction or measured evidence. More shovels alone do not ensure more throughput.

PPE set: 1 pair work gloves + 1 pair safety glasses per worker; closed-toe shoes required. Cards include one basic set per listed worker unless stated otherwise. Ordinary PPE is not hazardous-material protection.

Sandbags: counted empty bags; no precise bag weights.

Fill: cubic yards; 75 completed bags per yd³. Stock supports about fill × 75 bags.

Movement & deliveries

Wheelbarrow: 4 bags per trip, one operator, 10 min full load/travel/unload/return cycle on a dry level route up to 300 ft one way; 20 min for >300–600 ft. Use fewer bags or a buddy if needed; never force the stated load.

Utility cart: 8 bags per trip, one operator; same cycle times as wheelbarrow. Only approved smooth level routes. Hill, steps or poor ground require a different plan.

Pickup: one qualified adult driver; classroom load 40 bags OR 0.5 yd³ loose fill per trip, never both; 30 min load/travel/unload/return cycle on an approved dry vehicle route. Loading helpers come from the workforce.

Dump truck: supplier driver; 4 yd³ per delivery, 1 h per delivery cycle once service begins. It delivers fill, not finished barriers. Card-specific access and delays override the catalog.

No capacity calculation licenses vehicle overloading; real operators verify ratings. Adult classroom scenarios use these simplified bag counts instead of bag weights.

Trips = round UP (bags ÷ capacity). Transfer time = trips × cycle minutes ÷ 60; divide across identical vehicles only if independently staffed. Loose-fill pickup trips = round UP (yd³ ÷ 0.5).

Other resources you may request

Tarps/plastic sheets and compatible tape: record dimensions and overlaps; check the supplied coverage. Basin liner: base plus berm sides, freeboard and slack; leaks and seams matter. Filter fabric: scenario-specific erosion protection, not structural reinforcement.

Traffic cones and caution tape: mark work zones without blocking exits; never replace proper road closure authority. Rated signs/frames/tents and their required anchors: adult-approved use only.

Flashlights/work lights: request for darkness and verify power/fuel and electrical safety. Radios: plan 1 per team leader plus incident lead if available.

Drinking-water cooler: 5 gal capacity; safe refill source and cups/bottles needed. First-aid kit: at least 1 accessible per work site with trained adult contact.

Basic hand tools: closure supplies, tape measure and approved bag-holding aid; never improvise unsafe shovel stands. Shade/rest space, food, sanitation and waste collection must be arranged.

Scenario resource requests

No worker, bag, fill, equipment or PPE allocation is supplied on revised cards. Calculate and request each quantity, including drinking water, food, first aid, closure supplies and safe staging. State delivery and arrival assumptions. Equipment-specific approved ballast counts describe the mission requirement, not supplied stock.